INTELLIINSPECT AI

Industrial Vision & Automated QA System

CERTIFICATE: REP-20261031-0009
DATE: 2026-10-31 22:43:53 UTC
STANDARD: ISO 9001:2015 / ASTM E155

INSPECTION DECISION: REJECT

Status: DEFECTIVE | Severity: HIGH | Confidence: 82.3%

Product Code:	TEST-CI-001	**Machine ID:**	DC-CELL-01
Component Type:	Pump Impeller (Cast Iron/Alloy)	**Batch Number:**	B2026-01
Inspection Model:	PatchCore ResNet-18 (512x512)	**Inspector:**	IntelliInspect AI Vision Station #1

Visual Defect Localization & Anomaly Map

Figure 1: As-Received Casting Surface

Figure 2: AI Thermal Anomaly Map & Bounding Box

Quantitative AI Measurements

Parameter	Measured Value	Standard Limit	Compliance
Anomaly Peak Score	2.1681	Threshold: 1.6577	FAIL
Model Confidence	82.3%	>= 70.0%	PASS
Defect Classification	Molding Burr / Flash	None (Defect-Free)	FAIL
Severity Rating	HIGH	NONE / LOW (Acceptable)	FAIL

Smart Decision Explanation & Engineering Rationale

Engineering Notes: Critical defect identified (Molding Burr / Flash). Peak anomaly intensity 1.31x above tolerance limit. Rejection mandated to prevent catastrophic structural failure.

Digital Verification:

Scan QR code with any mobile device to verify certificate authenticity on the blockchain/cloud traceability registry.

Quality Assurance Sign-Off:
Lead QA Inspector: *A. Sharma, PE*
Station: *Die-Cast Cell #2*
Status: **VERIFIED RECORD**